EXP 10: Write a Python Program to Implement Expectation & Maximization Algorithm

CODING:

```
import math

# Input Dataset
X = [1, 2, 3, 8, 9, 10]

# Initialize Parameters
means = [2.0, 9.0]
variances = [2.0, 2.0]
weights = [0.5, 0.5]

# Gaussian Probability Function
def gaussian(x, mean, variance):
  return (1 / math.sqrt(2 * math.pi * variance)) * math.exp(
    -((x - mean) ** 2) / (2 * variance)
  )

# EM Algorithm
for iteration in range(20):

  # E-Step: Calculate Responsibilities
  responsibilities = []

  for x in X:
    p1 = weights[0] * gaussian(x, means[0], variances[0])
    p2 = weights[1] * gaussian(x, means[1], variances[1])
```

```
        total = p1 + p2

        r1 = p1 / total
        r2 = p2 / total

        responsibilities.append([r1, r2])

    # M-Step: Update Parameters
    for k in range(2):

        total = sum(r[k] for r in responsibilities)

        means[k] = sum(
            responsibilities[i][k] * X[i]
            for i in range(len(X))
        ) / total

        variances[k] = sum(
            responsibilities[i][k] * (X[i] - means[k]) ** 2
            for i in range(len(X))
        ) / total

        variances[k] = max(variances[k], 0.000001)

        weights[k] = total / len(X)

# Display Results
```

```
print("Input Data:", X)

print("\nFinal Cluster Means:")
print(means)

print("\nFinal Variances:")
print(variances)

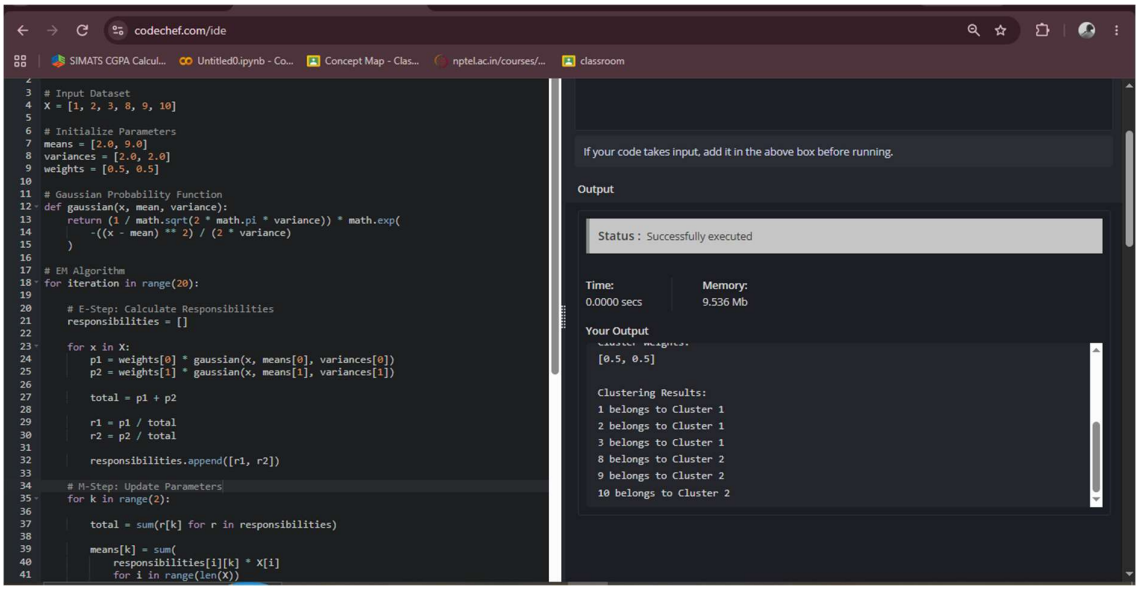
print("\nCluster Weights:")
print(weights)

# Assign Clusters
print("\nClustering Results:")

for x in X:
    p1 = weights[0] * gaussian(x, means[0], variances[0])
    p2 = weights[1] * gaussian(x, means[1], variances[1])

    if p1 > p2:
        print(x, "belongs to Cluster 1")
    else:
        print(x, "belongs to Cluster 2")
```

OUTPUT:


codechef.com/ide
SIMATS CGPA Calcul...
Untitled0.ipynb - Co...
Concept Map - Clas...
nptel.ac.in/courses/...
classroom
Input Dataset
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Gaussian Probability Function
def gaussian(x, mean, variance):
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EM Algorithm
for iteration in range(20):
E-Step: Calculate Responsibilities
responsibilities = []
for x in X:
p1 = weights[0] * gaussian(x, means[0], variances[0])
p2 = weights[1] * gaussian(x, means[1], variances[1])
total = p1 + p2
r1 = p1 / total
r2 = p2 / total
responsibilities.append([r1, r2])
M-Step: Update Parameters
for k in range(2):
total = sum(r[k] for r in responsibilities)
means[k] = sum(
responsibilities[i][k] * X[i]
for i in range(len(X))
If your code takes input, add it in the above box before running.
Output
Status : Successfully executed
Time:
0.0000 secs
Memory:
9.536 Mb
Your Output
[0.5, 0.5]
Clustering Results:
1 belongs to Cluster 1
2 belongs to Cluster 1
3 belongs to Cluster 1
8 belongs to Cluster 2
9 belongs to Cluster 2
10 belongs to Cluster 2